VvR

Market Risk Engine for Multi-Asset VaR, Expected Shortfall, and Regime Detection

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GitHub: https://github.com/ValVal-Data/VaR
Focus: Market risk modelling, VaR/ES, GARCH, Monte Carlo simulation, regime detection

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Project Summary

This project implements and validates a market risk engine for a diversified multi-asset portfolio. The objective is to estimate 1-week 97.5% Value at Risk (VaR) and Expected Shortfall (ES) under realistic return dynamics.

The framework compares historical simulation, parametric models, and Monte Carlo simulation. It evaluates both constant-volatility and time-varying volatility assumptions, as well as Gaussian, Student-t, and skewed-t innovation distributions.

The project is structured as a model selection and validation exercise. Each specification is assessed using distribution diagnostics, heteroskedasticity tests, dependence diagnostics, Monte Carlo convergence analysis, stress testing, and VaR backtesting.

Within the tested model universe and sample period, CCC-GARCH(1,1) with skewed-t innovations provides the strongest empirical fit, the most robust backtesting performance, and the most credible portfolio tail-risk estimates.

As an extension, a variational autoencoder was trained on rolling-window market features. The resulting two-dimensional latent space was used for unsupervised market-regime detection and stress monitoring. The VAE identified three statistically distinct market regimes.

Objective

The objective of this project is to build a reproducible market risk framework for estimating portfolio-level VaR and ES across multiple asset classes.

The focus is not only on producing a risk number, but on evaluating which modelling assumptions are empirically justified. The project therefore combines risk estimation, statistical validation, backtesting, and stress analysis in one workflow.

The project demonstrates end-to-end quantitative risk-model development, including:

- Data preparation and portfolio construction
- Return distribution analysis
- Volatility modelling
- Dependence modelling
- Monte Carlo simulation
- VaR and ES estimation
- Model validation and backtesting
- Stress testing
- Unsupervised regime detection using deep learning

Executive Findings

- Empirical returns are non-Gaussian, asymmetric, and heavy-tailed.
- Volatility is time-varying across the sample, making constant-volatility models inadequate for current risk measurement.
- Heavy-tailed and asymmetric innovation distributions materially improve portfolio tail-risk estimation.
- CCC-GARCH(1,1) with skewed-t innovations provides the strongest overall validation results among the tested specifications.
- Portfolio tail risk is more sensitive to volatility shocks than to correlation shocks.
- The ES/VaR ratio remains relatively stable under stress scenarios, indicating that stress mainly scales loss magnitude rather than materially changing tail shape.
- The VAE extension identifies three statistically distinct market regimes from rolling-window risk features and provides a complementary stress-monitoring layer.

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Data and Portfolio

The analysis uses weekly data over a five-year period for a diversified multi-asset portfolio.

The portfolio includes exposure to:

- Global equities
- Swiss real estate
- Gold
- Global corporate bonds
- Foreign exchange exposure

The investable universe consists of the following instruments:

- iShares MSCI ACWI, USD, accumulating
- UBS SXI Real Estate, CHF, distributing
- UBS Gold, USD, distributing
- iShares Global Corporate Bond, CHF, distributing
- USD/CHF exchange rate
- EUR/CHF exchange rate

The portfolio is designed to provide broad cross-asset exposure. Portfolio weights are defined externally in the project configuration file, allowing the engine to be reused with different allocations.

Risk Measures

- The analysis estimates 1-week 97.5% VaR and Expected Shortfall.
- VaR is defined as the negative 2.5th percentile of the 1-week portfolio profit-and-loss distribution.
- Expected Shortfall is defined as the average loss conditional on losses exceeding the VaR threshold.

VaR provides a loss threshold at a specified confidence level. Expected Shortfall provides additional information on the severity of losses beyond that threshold and is therefore more informative for tail-risk assessment.

Model Universe

The project benchmarks the following classes of risk models:

- Historical simulation
- Parametric models
- Monte Carlo simulation

The implemented specifications are:

- Historical simulation

- Gaussian parametric model
- Exponentially weighted moving average model
- Geometric Brownian motion with historical volatility
- CCC-GARCH(1,1) with Gaussian innovations
- CCC-GARCH(1,1) with Student-t innovations
- CCC-GARCH(1,1) with skewed-t innovations

The model set is designed to test progressively richer assumptions: from simple historical and Gaussian models to time-varying volatility models with heavy-tailed and asymmetric innovations.

Model Selection Approach

Model selection is based on empirical adequacy rather than model complexity alone.

Each model is assessed using:

- Distribution diagnostics
- Normality testing
- Heteroskedasticity testing
- Dependence diagnostics
- Residual analysis
- Monte Carlo convergence checks
- VaR backtesting
- Stress testing

The final model is selected based on statistical performance, backtesting quality, and economic interpretability.

Within this framework, CCC-GARCH(1,1) with skewed-t innovations is selected as the final specification.

Distributional Results

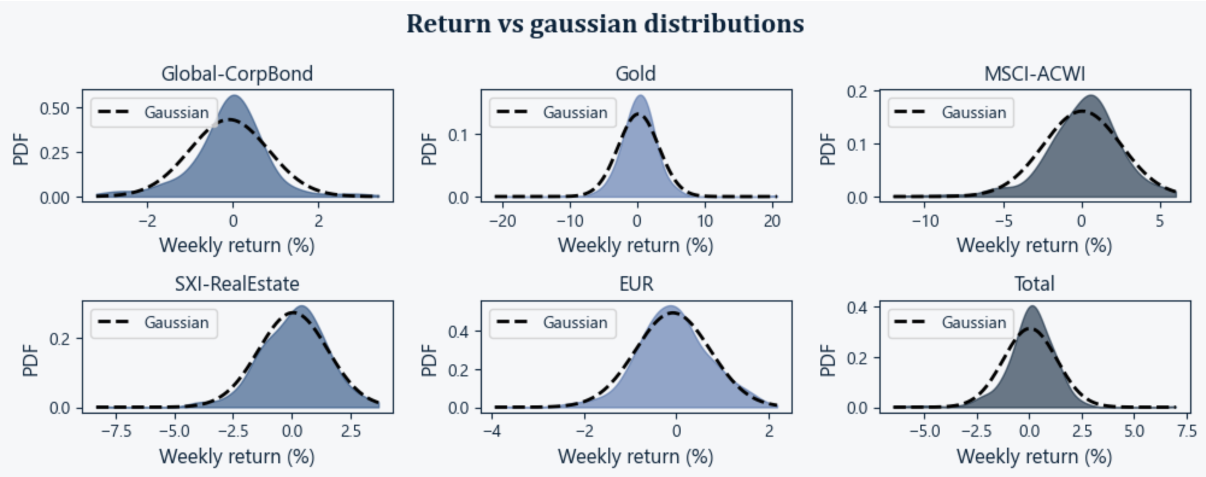

Figure 1 Empirical asset return distributions, with fitted Gaussian densities shown as dashed lines.

As shown in Figure 1, the empirical return distributions are not Gaussian. They show asymmetry, excess kurtosis, and heavy tails.

This supports the use of innovation distributions that can capture both fat tails and skewness. The skewed-t specification is suitable because it combines the tail flexibility of the Student-t distribution with an asymmetric shape parameter.

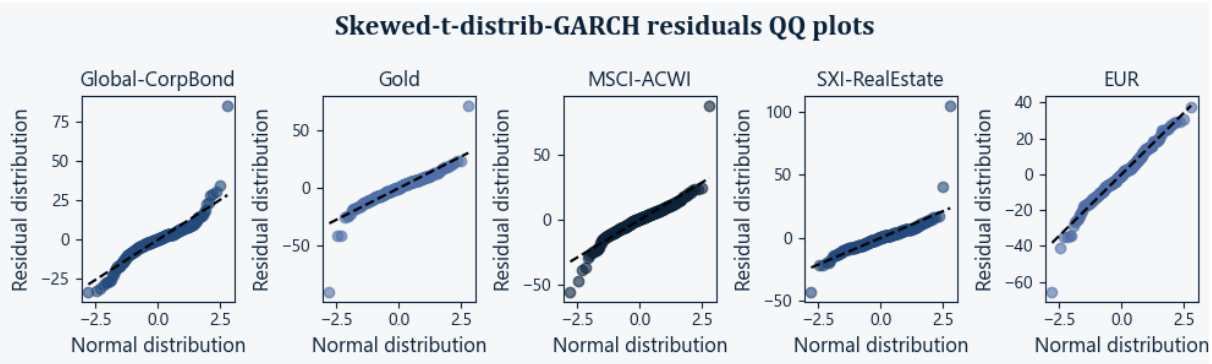

Figure 2: Residual Q-Q plots against the Gaussian reference distribution.

Figure 2 shows that GARCH-based specifications materially improve model fit relative to simpler models. Allowing for heavy-tailed and asymmetric innovations produces residual behaviour that is more consistent with the empirical return distribution.

VaR and Expected Shortfall Results

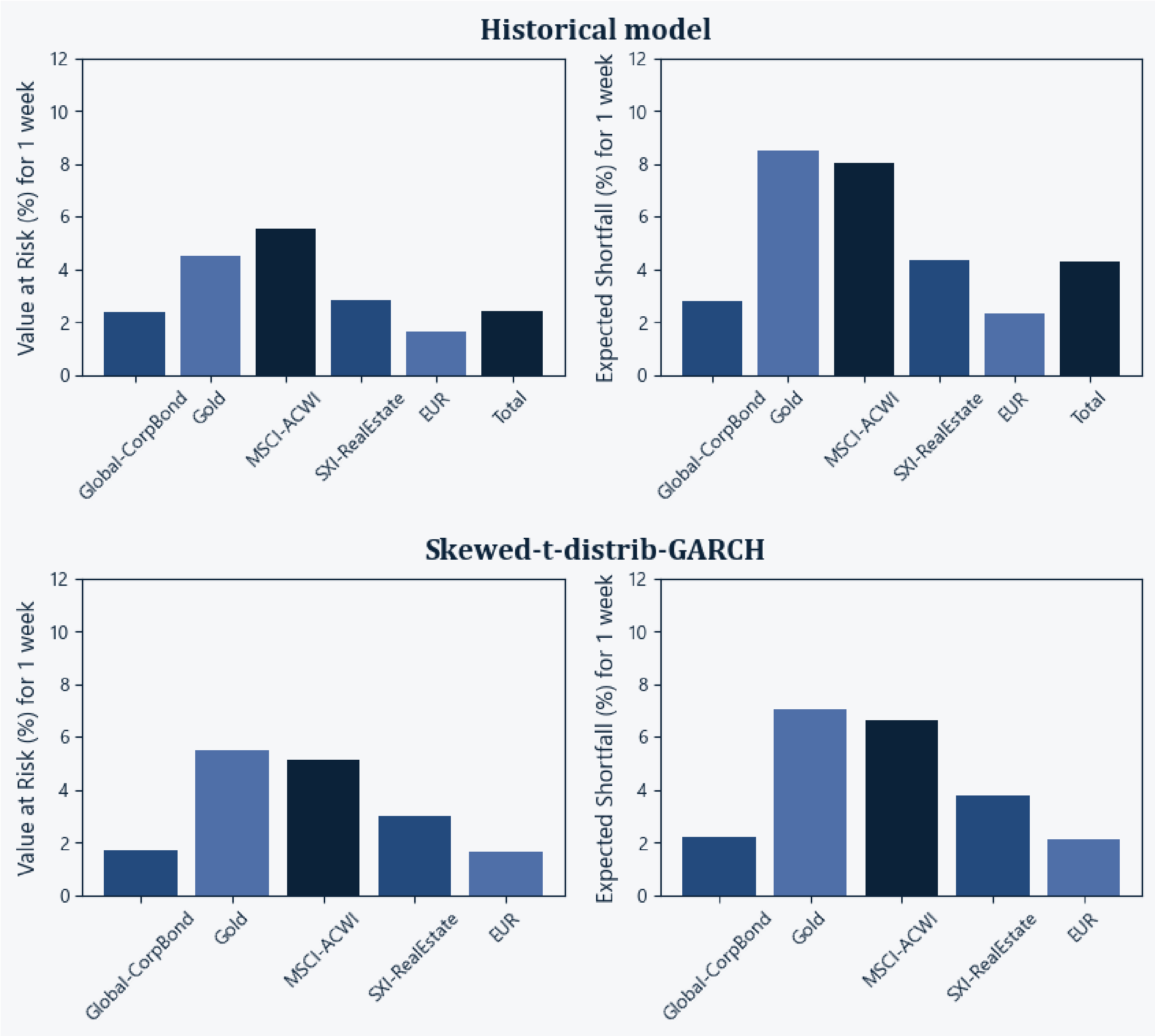

Figure 3: Historical simulation and GARCH-based portfolio loss distributions with VaR and ES estimates.

The comparison between historical and Monte Carlo-based risk estimates in figure 3 shows that volatility-sensitive models respond more strongly to current market conditions than long-window historical simulation.

This is economically intuitive. Historical simulation pools calm and stressed periods in the same empirical distribution and may underreact when the current volatility regime changes.

By contrast, GARCH-based Monte Carlo simulation updates the conditional volatility estimate and is therefore more responsive to recent market dynamics.

The selected skewed-t CCC-GARCH specification provides the most credible 1-week VaR and ES estimates among the tested models.

Monte Carlo Convergence

Monte Carlo convergence is assessed by comparing simulated return distributions and tail-risk estimates across increasing numbers of simulation paths:

- 100 paths
- 1,000 paths
- 10,000 paths
- 100,000 paths

As the number of simulation paths increases, both the simulated return distribution and the estimated tail quantiles stabilize (See Figure A1 in Appendix).

Convergence appears acceptable from approximately 1,000 to 10,000 simulations. The final implementation uses 100,000 paths to reduce Monte Carlo noise in extreme quantiles and improve the stability of Expected Shortfall estimates.

Backtesting

VaR backtesting is used to assess whether the selected model produces exception rates consistent with the target confidence level.

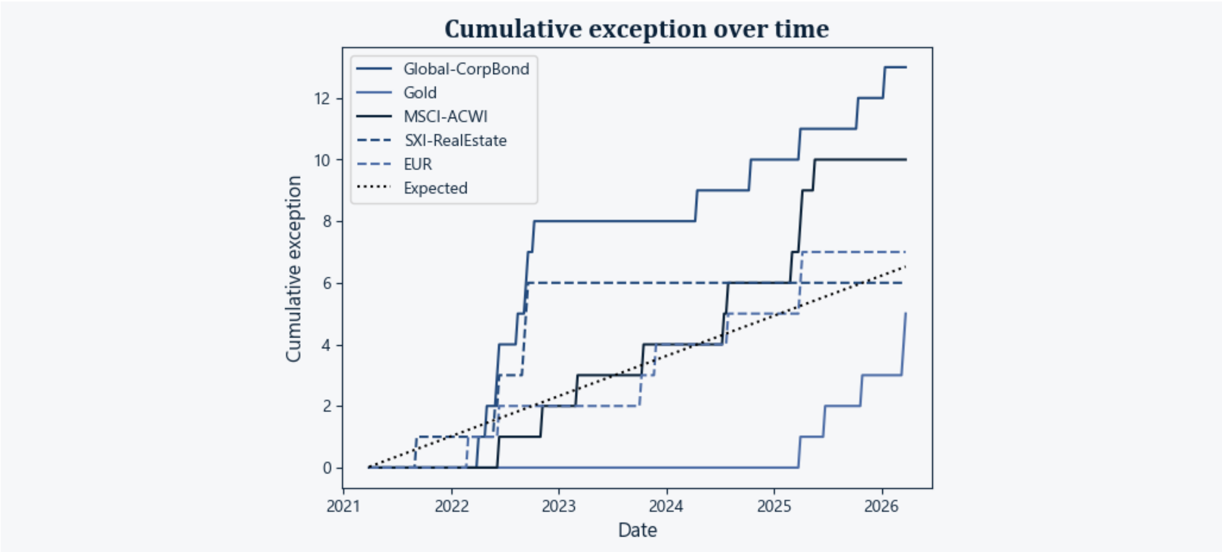

Figure 4: Cumulative VaR exception for the selected skewed-t CCC-GARCH specification

The selected CCC-GARCH(1,1) model with skewed-t innovations produces exception behavior that is consistent with the 97.5% VaR level over the sample considered as shown in Figure 4.

Backtesting supports the use of this specification for portfolio-level tail-risk estimation, subject to the stated model and sample limitations.

Stress Testing

Stress testing is used to assess the portfolio's sensitivity to volatility and correlation assumptions.

Table 1: VaR multipliers under correlation, volatility, and combined stress scenarios relative to the baseline model.

	Global-CorpBond	MSCI-ACWI	Gold	SXI-RealEstate	EUR
Model	1.0	1.0	1.0	1.0	1.0
Correlation stress	0.9	0.9	0.7	0.8	1.0
Volatility stress	2.6	3.0	3.3	2.7	2.6
Volatility+Correlation stress	2.6	3.1	3.4	2.5	3.5

Table 1 shows that volatility stress produces materially larger increases in VaR and ES than correlation stress. This indicates that portfolio tail risk is primarily driven by volatility shocks rather than by changes in cross-asset correlation.

Table 2: ES/VaR ratios under correlation, volatility, and combined stress scenarios.

	Global-CorpBond	MSCI-ACWI	Gold	SXI-RealEstate	EUR
Model	1.3	1.3	1.3	1.3	1.3
Correlation stress	1.0	1.3	1.3	1.4	1.1
Volatility stress	1.5	1.8	1.3	1.2	1.2
Volatility+Correlation stress	1.4	1.2	1.2	1.3	1.2

The ES/VaR ratio remains relatively stable across stress scenarios as shown in Table 2. This suggests that stress mainly scales the magnitude of losses rather than materially changing the shape of the loss tail.

This result is consistent with the linear structure of the portfolio. The portfolio does not contain strongly nonlinear instruments such as options.

Model Selection Summary

- Historical simulation is retained as a transparent benchmark. It is easy to interpret but slow to adapt to changing volatility regimes.
- The Gaussian parametric model is rejected because empirical returns are non-normal, asymmetric, and heavy-tailed.
- The EWMA model improves volatility responsiveness but does not fully address residual tail behaviour.
- The GBM specification with constant volatility is rejected because heteroskedasticity is present in the data.

- CCC-GARCH with Gaussian innovations captures volatility clustering but underestimates tail risk.
- CCC-GARCH with Student-t innovations improves tail fit and provides acceptable validation results.
- CCC-GARCH with skewed-t innovations provides the strongest overall performance within the tested model universe and is selected as the final model.

Selected Model

The selected final model is CCC-GARCH(1,1) with skewed-t innovations.

This specification captures three empirically relevant features of the data:

- Time-varying volatility
- Heavy-tailed shocks
- Asymmetric return innovations

The model provides a strong balance between statistical fit, economic interpretability, and implementation tractability.

Deep-Learning Extension: VAE-Based Regime Detection

The project extends the VaR framework with an unsupervised deep-learning module for market-regime detection.

A variational autoencoder is trained on rolling-window feature vectors derived from the portfolio time series. The input features are:

- Return
- Volatility
- Skewness
- Kurtosis
- Cumulative return
- Z-score

The objective is to compress these market features into a lower-dimensional latent representation and assess whether the latent space separates distinct market regimes.

VAE Methodology

The VAE architecture is optimized using random search over 10,000 model configurations. Poorly performing models are stopped early to reduce computational cost.

The best-performing architecture uses a two-dimensional latent space. This provides a useful balance between compression quality and interpretability, as the latent representation can be visualized directly.

The trained latent space is analyzed through:

- Latent-space displacement over time
- Latent-space clustering

Clustering performance is assessed using three internal validation metrics:

- Silhouette score
- Calinski-Harabasz score
- Davies-Bouldin score

The final clustering analysis identifies three latent regimes. These clusters are then compared against the original rolling-window features using ANOVA to assess whether the regimes are statistically distinct.

VAE Results

The VAE learns a structured two-dimensional latent space as shown in Figure 5. Market states do not move randomly in this representation. Instead, the portfolio transitions between distinct regions of the latent space, suggesting that the model captures regime-like market behavior.

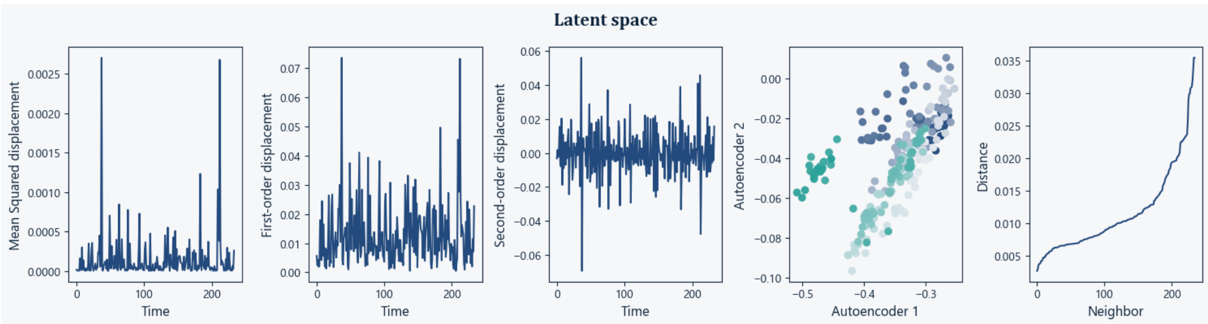

Figure 5: Latent-space diagnostics for the VAE regime-detection model. Panels show mean squared displacement, first- and second-order latent displacement, the two-dimensional latent space, and mean k-nearest-neighbour distance as a function of k.

The clustering analysis identifies three statistically distinct regimes as shown in Figure 6.

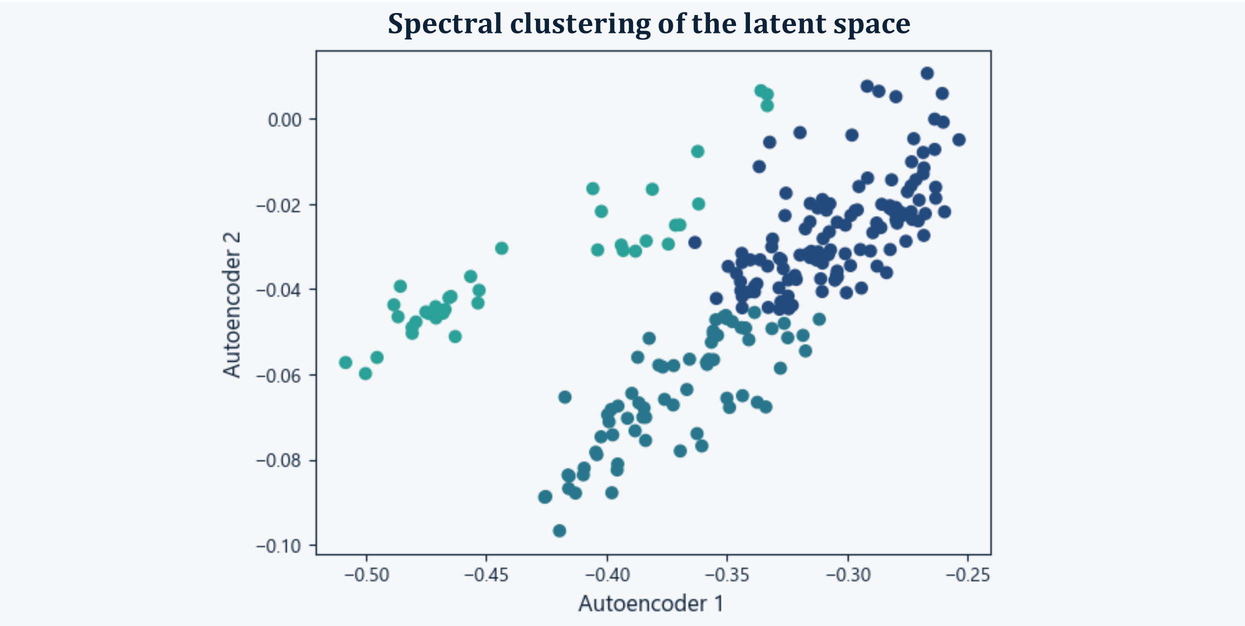

Figure 6: Two-dimensional VAE latent space coloured by the three identified market regimes.

The feature averages shown in Figure 7 suggest the following interpretation:

- Cluster 1: Weak market regime
  This regime is associated with weaker realised performance and limited upside. It resembles a flat or deteriorating market environment and may correspond to risk-off or slow-moving conditions.
- Cluster 2: Favourable market regime
  This regime is associated with stronger realised returns, comparatively contained volatility, and more favourable distributional characteristics. It represents the most favourable realised return-risk profile in the sample.
- Cluster 3: High-volatility stress or opportunity regime
  This regime is associated with elevated volatility, fatter tails, and larger market moves. It may include both sharp rallies and sharp drawdowns. Risk is materially higher in this regime.

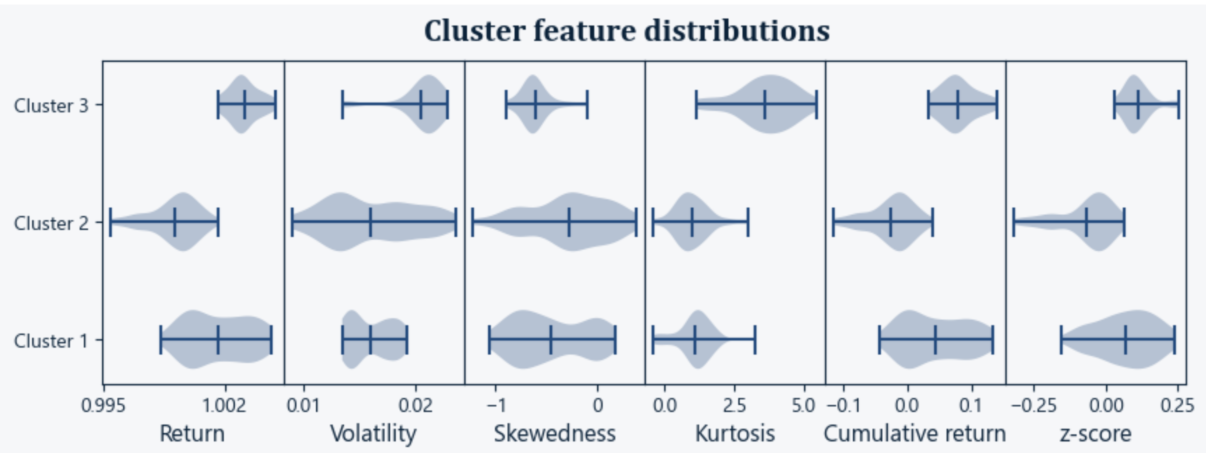

Figure 7: Distributions of rolling-window features across the three identified VAE regimes.

ANOVA confirms that the three clusters differ significantly across the original input features. This indicates that the latent-space separation reflects meaningful differences in market behaviour rather than arbitrary clustering.

Interpretation of the VAE Extension

The VAE-based regime analysis provides a complementary layer to the classical VaR and ES framework.

The GARCH model explicitly models volatility dynamics and distributional assumptions for tail-risk estimation. The VAE, by contrast, learns nonlinear interactions across multiple rolling-window features and maps them into a compact market-state representation.

The latent-space position and latent-space displacement can therefore be used as stress-monitoring indicators.

The VAE is not a replacement for VaR or Expected Shortfall. Its value is to provide an additional, non-parametric view of market-state transitions and potential stress conditions.

Model Limitations

The analysis is subject to several limitations:

1. Cross-asset dependence is modelled through a constant conditional correlation structure. This is tractable and transparent, but correlations may become time-varying during periods of acute market stress.
2. VaR and ES are model-based estimates. They depend on the selected volatility model, innovation distribution, calibration window, and portfolio composition.
3. The results are sample-dependent. Different calibration periods may lead to different parameter estimates, validation outcomes, and risk estimates.
4. The portfolio contains linear instruments only. Nonlinear instruments such as options are not included. Convex payoff effects and nonlinear tail amplification are therefore outside the current scope.
5. The VAE identifies regimes from historical rolling-window features. It does not yet provide a fully out-of-sample forecast of future stress regimes.
6. While the two-dimensional latent space improves interpretability, the VAE remains a nonlinear model. Economic interpretation of latent regimes still requires judgement.

Implementation

The project is implemented in Python.

The main libraries used are:

- NumPy
- SciPy
- Pandas
- scikit-learn
- PyTorch
- Matplotlib
- Jupyter Notebook

The implementation is modular, with separate components for:

- Data handling
- Distribution modelling
- Monte Carlo simulation
- Statistical testing
- Plotting
- Neural-network modelling
- Utility functions

The codebase is designed for reproducibility, extension, and independent validation.

Conclusion

This project implements a complete market-risk workflow for a diversified multi-asset portfolio.

The analysis shows that Gaussian and constant-volatility assumptions are insufficient for the observed return dynamics. Empirical returns are heavy-tailed, asymmetric, and heteroskedastic.

Within the tested model universe, CCC-GARCH(1,1) with skewed-t innovations provides the strongest combination of distributional fit, backtesting performance, and economic interpretability.

Stress testing indicates that portfolio tail risk is primarily driven by volatility shocks rather than correlation shocks.

The VAE extension adds an unsupervised regime-monitoring layer. It identifies three statistically distinct market regimes from rolling-window risk features and can be used as a complementary stress indicator alongside the VaR and ES engine.

Overall, the project combines classical market-risk modelling with modern representation learning in a validation-driven framework.

Future Extensions

Potential extensions include:

- Dynamic conditional correlation models
- Extreme value theory for tail modelling
- Out-of-sample regime prediction
- Inclusion of nonlinear instruments
- Option-based portfolio risk
- Interactive risk dashboard

Appendix

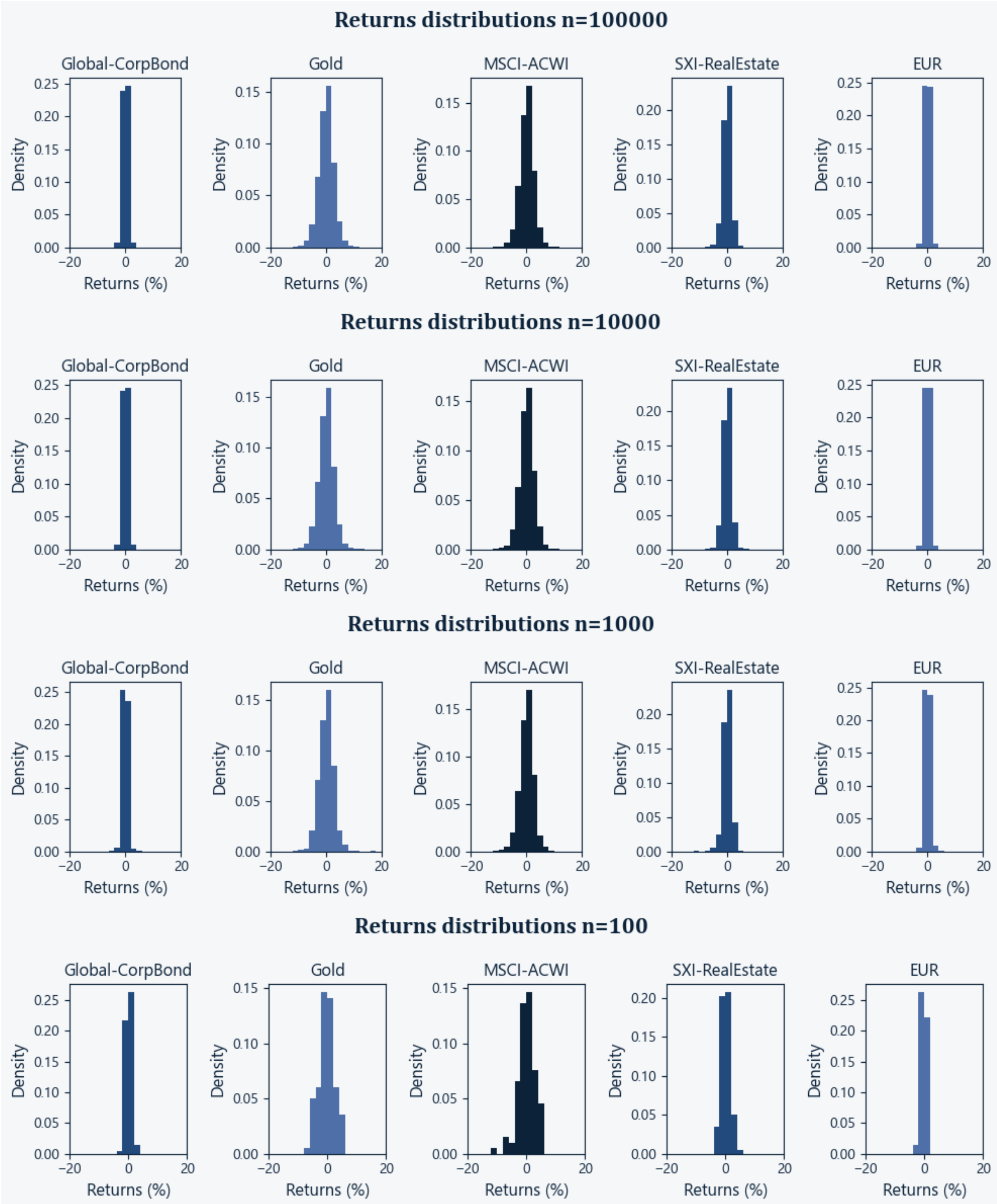

Figure A1: Monte Carlo portfolio return distributions across different numbers of simulation paths.